

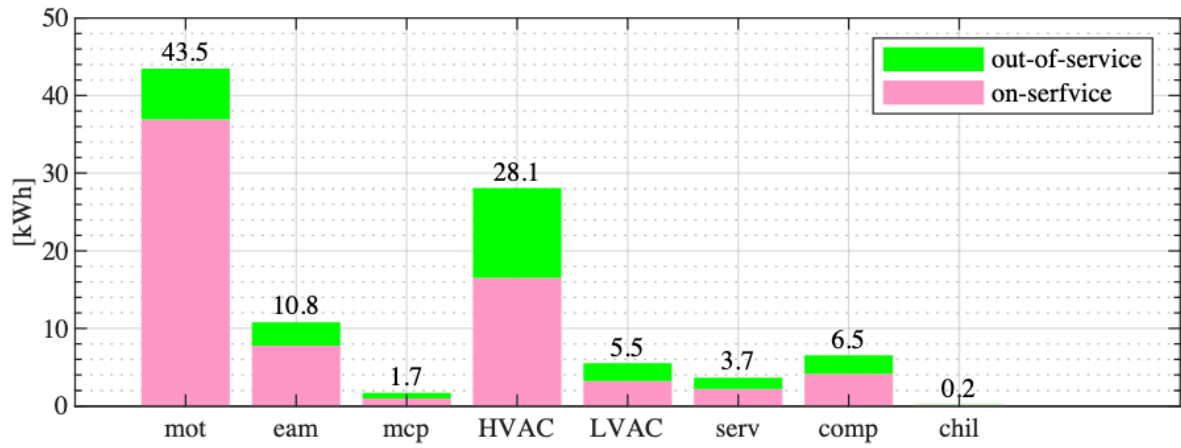


Figure 17. Repartition of the total energy consumed during the experimental tests.

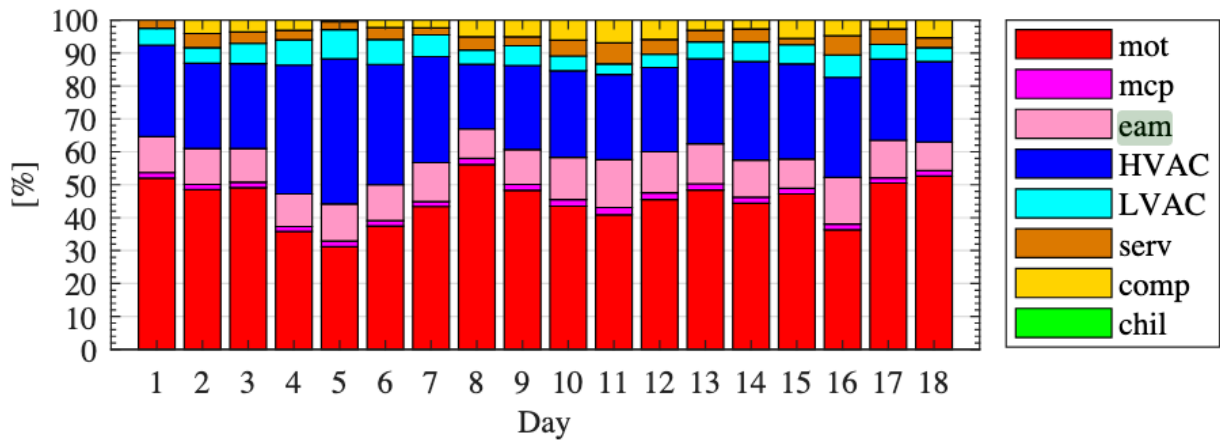


Figure 6. Energy consumption distribution of the systems of the electric bus.

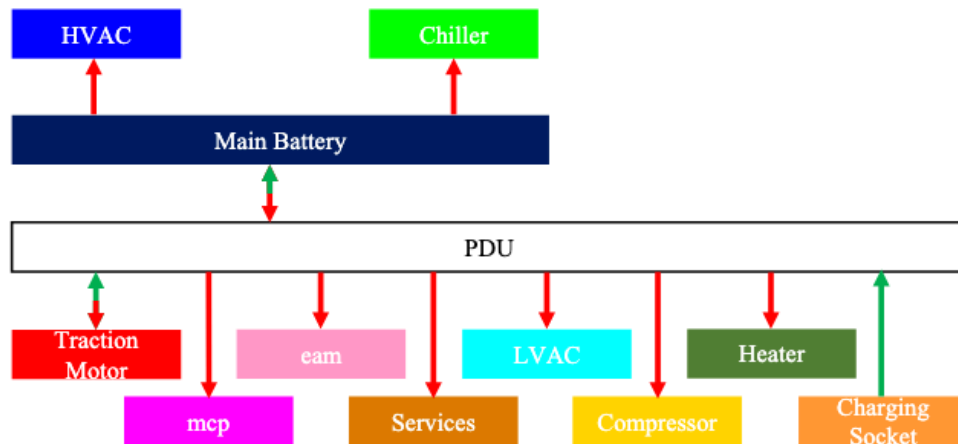


Figure 1. Electric scheme of battery-electric buses.

- “Traction motor”: This includes the energy consumption of the AC traction motor and the inverter. The available data include the voltage and current on the DC side of the inverter.
- “mcp”: This refers to the motor cooling pump. The available data pertain to the voltage and the current absorbed. Although it is part of the 24 V circuit, it is essential for vehicle movement.
- “eam”: This is the electric power steering.
- “LVAC”: This is the air-conditioning system powered by the low-voltage circuit, including the ventilation fans and the driver’s cabin air-conditioning.

- “Services”: This refers to the power consumed by the 24 V circuit, excluding “MCP” and “LVAC”. This includes the power used by vehicle lights, door-opening mechanisms, and similar services.
- “Compressor”: This refers to the power consumed by the compressor in the suspension system.
- “Heater”: This is used to heat the vehicle cabin during winter but was not considered in this study due to summer measurements.
- “Chiller”: This is the battery pack chiller.
- “HVAC”: The high-voltage air-conditioning system cools the bus cabin. Like the battery chiller, this utility draws power directly from the battery pack.
- “Battery”: The current and voltage at the battery pack terminal were monitored to ensure the sum of energy consumed by all peripherals matched the power output from the battery pack.